SKYVIEW WAGA DISPLAY

USER AND INSTALLATION MANUAL (FOR VERSION 3)

Please read this manual carefully before using the device. This manual is an essential component of the device and should be kept in a safe place so that it may be referred to when required.

This manual is specific to the version 3 of the SkyView WAGA Display.

Never fly with a SkyView display without extensive familiarisation on ground. Train on the ground, before flying, on how to interpret and react in case of a Flarm Alarm warning. Observe restrictions and safety instructions detailed in this manual

WARNING AND CONDITIONS OF USE

The pilot is ultimately responsible for all flight decisions and for operating the aircraft safely at all times.

Pilots must maintain good visual lookout to avoid collision.

The SkyView WAGA Display can only display traffic equipped with a limited range of electronic conspicuity device types and electronic systems are fallible, therefore the display may not show all aircraft in the proximity and must not be relied upon as a traffic warning device.

For situational awareness only. Never make safety critical decisions based on displayed information.

No liability is accepted by the producers of the SkyView WAGA Display or its software.

Record of changes

Version	Date	Description
WAGA01	1/5/25	Initial release. Enhancements to the standard Soft RF SkyView include: Redefined use of nav boxes. Landscape mode. PCAS displayed. Added alarms displayed on radar with relative bearing lookout indicator. Added Nav_View with bearing and distance to a selected Waypoint. For tugs doing paddock retrieves or emergency navigation back to base.
WAGA02	1/2/26	Added CompID shown beside traffic icon on radar. Alarm screen now obscures radar until alarm stops. Added RMI needle and back bearing displayed on Nav_View screen radar. Reassigned order and titles of some of the nav boxes.
WAGA03	13/04/26	Added vertical angle indicator on alarm screen. Blocked CompIDs being shown for traffic with zero ground speed, to eliminate clutter at launch line up. Does display in Mode_Nav to aid towplanes to find glider in paddocks. Tug CompID now shown at 2 o'clock to traffic icon and all other aircraft at 4 o'clock. Enables both IDs of a tug owing a glider to be easily read. Added option 'TONE" ' to Sounds setting. This plays a .wav sound file to the speaker/headset for each Advisories/alarms instead of the composite o'clock/distance/altitude message provide by option VOICE. Fixed issue of advisory filters 'Altitude +/-500 m' and 'Alarm only' blocking display of all traffic above or below 1640ft.

Terminology used in this manual:

EC	Electronic Conspicuity. An aviation electronic device that transmits and receives radio signals with the intention of improving situational awareness and collision avoidance. Eg FLARM, ADSB, Mode-S, SoftRF. Comprises a transceiver and a display, which may be integrated but are more usually separate components.
FLARM	An aviation EC system. The registered trademark of FLARM Technology AG.
FLARM Alarms	Alarm levels assessed by FLARM EC devices giving a warning of time to impact. Defined by FLARM.
Initial batch	The May 2025 batch of WAGA Flarm and SWD, approximately 35 of each.
NMEA	National Marine Electronics Association standard protocol for data encoding.
PowerFlarm	The processor chip set with FLARM propriety software which is built into different manufacturers EC devices. Successor to the ‘Classic’ FLARM chip set.
Rx/Tx	Receive/Transmit
SoftRF	A “DIY, multi-functional, compatible, sub-1 GHz ISM band radio-based proximity awareness system for general aviation”. Open-source software.
SkyView	An aviation electronic conspicuity display from the SoftRF open-source project
SWD	SkyView WAGA Display. A development of the SoftRF SkyView EZ. Subject of the manual. Produced not-for-profit.
ThisAircraft	The aircraft in which the SWD is installed.
Threat	The traffic with the highest level of Flarm Alarm or, if no Flarm Alarms, then the closest traffic.
Traffic	Aircraft detected and available for display on the SWD.
WAGA	Western Australia Gliding Association (WAGA) Inc. Producers of the SWD and WAGA Flarm.
WAGA Flarm	An EC transceiver based on the PowerFlarm Atom. Produced not-for-profit.
WebUI	Web User Interface, the application used on a smart phone to configure the SWD

Acknowledgements

A comprehensive acknowledgement list can be found on the Github references above or the WebUI.

Contents:

1 General

1.1 System status

The SkyView WAGA Display (SWD) is a development of the SoftRF EZ SkyView and is an open-source project, meaning:

- It is a development by multiple authors and designers.
- It is not a commercial product and must not be produced or sold for financial profit.
- No warranty is provided and no liability accepted by the producer.
- Technical support may be limited or not available from the producer.
- All information required to produce, maintain or develop it, on a DIY basis, is publicly available.

WAGA is the Western Australia Gliding Association. The SWD is the WAGA version of the SoftRF EZ SkyView. It was developed in January 2025 by and for the use of its members as a low-cost display, primarily to be coupled with the WAGA Flarm - a not-for profit PowerFlarm based EC transceiver.

The SWD is an Experimental Device, subject to continuing evaluation and improvement.

1.2 System Description

1.2.1 SkyView EZ Display

The SkyView EZ is a small and affordable aircraft cockpit display for traffic conspicuity information, with designs originated by Linar Yusupov as part of the SoftRF open-source project.

SkyView EZ shows traffic information provided by a FLARM NMEA or Garmin GDL90 data source device.

SkyView can display traffic data from any device that passes suitable data in the NMEA Flarm sentence formats, plus valid NMEA GPS fix format sentences. Data sentences must end with a valid Error Checksum or are discarded.

When connected to a suitable EC transceiver functioning correctly, the system:

1. Highlights alarm warnings by buzzer or voice (best in headset cockpits) and by highlighting the Threat aircraft on the display to assist lookout.
2. Aids situation awareness of other traffic by presenting a radar like picture of detected traffic within the selected zoom level.

The system does not provide traffic avoidance instructions.

The system does not provide obstacle warnings or airspace warnings.

Data input to the SkyView can be via a hard-wired serial interface, WiFi UDP connection, Bluetooth SPP or Bluetooth LE.

The SkyView EZ is only a display device therefore it must be paired in ThisAircraft with a working EC transceiver. Examples of types of suitable aviation EC transceivers for pairing with the SkyView EZ in a glider are: SoftRF, Classic Flarm, PowerFlarm, WAGA Flarm, SkyEcho (GDL90 device).

1.2.2 Electronic Conspicuity Transceivers

EC systems rely on aircraft having transceivers (a combined transmitter and receiver) that communicate with similarly equipped aircraft within range. Most systems use GPS to encode an aircraft's own position which they broadcast to other aircraft and receive similar broadcast data from other aircraft. This data can be displayed and may be processed to identify potential collision threats.

The three main systems of aviation EC transceivers in use are:

- FLARM - Widely used by gliders and glider tugs
- ADSB - Increasing used by General Aviation (GA) small aircraft
- Mode-S - This is secondary radar. Used by General Aviation and commercial aircraft.

NOTE: SoftRF mimics FLARM and provides FLARM NMEA data sentences but may not match all FLARM's capabilities.

Unfortunately, these three systems cannot communicate directly with each other. There are EC transceivers that have dual or multiple system capability. For example a PowerFlarm may have an ADSB processor built in which provides for ADSB-IN only (no ASDB-OUT) and Mode-S reception.

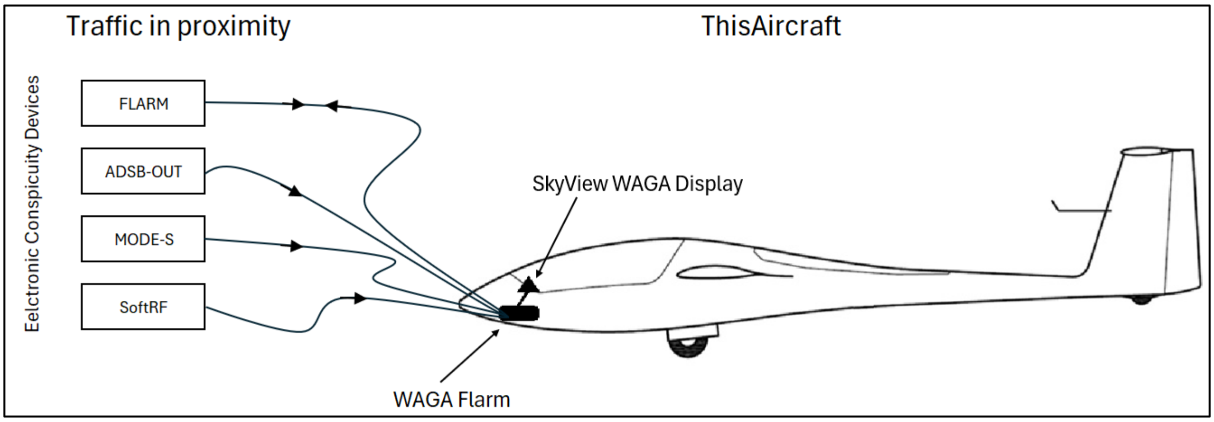


Illustration: Traffic in the proximity equipped with EC devices sending/receiving data with ThisAircraft

An EC transceiver usually processes the signals it receives from the traffic's transceivers and may filter out some data. It is essential to understand how ThisAircraft's EC transceiver may be limiting the traffic data passed to the SkyView display. For example, a PowerFlarm may be set to only show traffic within a specified range, or the selection of the PowerFlarm data port protocol may turn-off a data stream or limit the fields of data passed. Refer to the EC transceiver's manual.

For traffic to be detected, they must be in range of ThisAircraft's transceiver. The range achieved depends on the transmit power output, reception sensitivity and is influenced by the altitude of the aircraft and obstacles blocking radio signal 'line of sight'.

Thus the key factors are:

- Correctly configure ThisAircraft's EC transceiver
- Ensure ThisAircraft's antenna performance is good

1.2.3 SkyView WAGA Display

The SWD was developed from the SkyView EZ. It is designed for use with the WAGA Flarm and to meet the WAGA members' needs. There is nothing preventing its use by others.

The SWD displays traffic and alarms differently to the SkyView EZ and there are additional features.

A key difference is how the SWD displays FLARM alarms and information that warn of likely collision. The intent is to prompt the pilot to quickly lookout in the direction of the threat so he/she can scan a segment of the sky to identify the threat and take evasive action if needed.

A basic navigation feature has been added to assist aero retrieves of gliders or as a backup navigation device.

The SWD has been designed for partnering with a PowerFlarm EC transceiver (preferably with ADSB-IN) and inter-connection via a hardwired serial data and power cable. Connectivity to other EC transceivers and via non-hardwired connection has been retained but not tested.

1.3 System components

The SWD comprises a 85 x 50 x 18mm plastic case with a 2.7" (diagonal measurement) screen. The case encloses a printed circuit board with integrated E-Paper screen and space for a battery. The circuit board runs a software program derived from SoftRF written in the C++ coding language.

1.3.1 Software

The SWD uses a development of the SoftRF SkyView EZ software code originated by Linar Yusupov (copyright © 2016-2025). He describes SoftRF as a "DIY, multi-functional, compatible, sub-1 GHz ISM band radio-based proximity awareness system for general aviation". He has written code and produced designs for a family of traffic continuity receivers and displays. He has generously made the software available as open-source software on the website GitHub, subject to a licence which applies to the SkyView WAGA Display software and developments thereof, as below.

The original author of the SoftRF software has given the following licence:

(ref https://github.com/lyusupov/SoftRF/blob/master/software/firmware/README.md as at 29 March 2025)

Revision 0.12 of the SoftRF SkyView software was modified by Moshe Branner to provide extra functionality. His developments and code can be found on Github at:

https://github.com/moshe-braner/SoftRF/tree/master/software/firmware/binaries/ESP32/SkyView

The SWD software version 3.0 has been further developed by a WAGA member from the Moshe Branner version MB07C. Key features introduced in the SWD include:

- Additional choice of Landscape screen orientation, increasing installation flexibility.
- Addition of a selectable Nav_View screen with GPS calculated bearing and distance to a small list of waypoints (eg local airfields) to aid gliders reporting outlanding and tugs undertaking aero retrieves. Also has RMI needle that points to the waypoint.
- FLARM alarm 'look-out' indicator with vertical angle indicator replaces radar screen when alarms are triggered.
- Parsing of the 'Source' field from the $PFLAA string, enabling ADSB and Mode-S targets to be identified as such (requires FLARM protocol 9 or higher).
- Distinctive representation on the radar of gliders, tugs, ADSB and Mode-S traffic.
- More relevant and useful information shown in Nav Boxes.
- Comp IDs shown beside traffic icons on radar screen

The SWD software and notes are available at: https://github.com/Bumpff/SoftRF_Skyview.

1.3.2 Circuit board and screen

The circuit board is the LILYGO® TTGO T5S V2.8 ESP32 with 2.7 Inch E-Paper screen. This is a Chinese commercial component readily available on Amazon. Other than loading the SkyView WAGA Display software and adding a buzzer, no modifications are required to the board.

This component may be subject to unannounced changes by the manufacturer and there is no certainty that this will remain in production.

The SWD uses an E-Paper screen showing black text or graphics on a white background, mimicking the appearance of ordinary ink on paper. They use a different technology to conventional LCD screens and rely on ambient light to be read. One characteristic of E-Paper is that they continue to display the last image prior to power falling below a threshold level (eg battery flat or master-switch off when no battery is fitted. This can make a system appear frozen.

There is no luminance adjustment available.

1.3.3 Case/enclosure

The 2-part case is produced by 3D printing in plastic. There are currently no known sellers of a pre-printed case. DIY or contracted bespoke production is required using STL files found at:

https://github.com/lyusupov/SoftRF/tree/master/case/SkyView

The SWD uses a modified version of the case printed in nylon plastic.

1.3.4 Battery

The SoftRF EZ SkyView is designed to have a 3.7V battery within the case. When hardwired to a Flarm device, the SWD is powered by the aircraft power system therefore the internal battery is not required. An internal or external battery may be installed if desired.

1.3.5 Buzzer

A small buzzer is connected to the circuit board and located within the case. The buzzer may have an irregular sound because of the way the program turns it on and off so as not to add delays in the program loop. It is not Morse code!

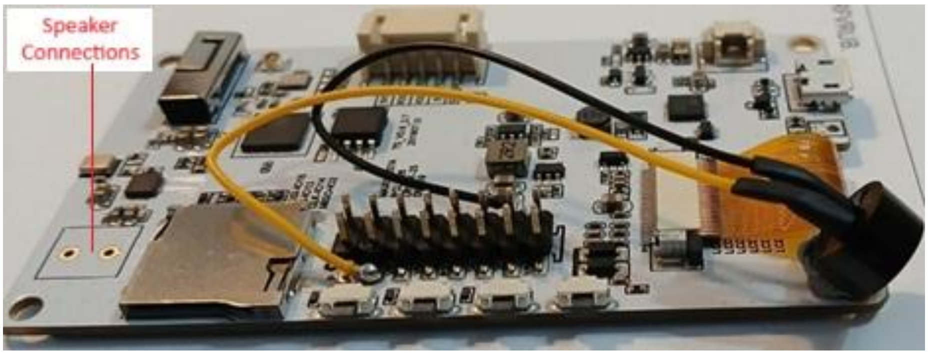


Illustration: Buzzer fitted as standard in SWD

2 Physical controls and connections

2.1 Control buttons

There are 4 press buttons on one long side of the case. On the SWD version of the case the two centre buttons are raised and the outer buttons flat. This is to allow tactile identification and use without looking.

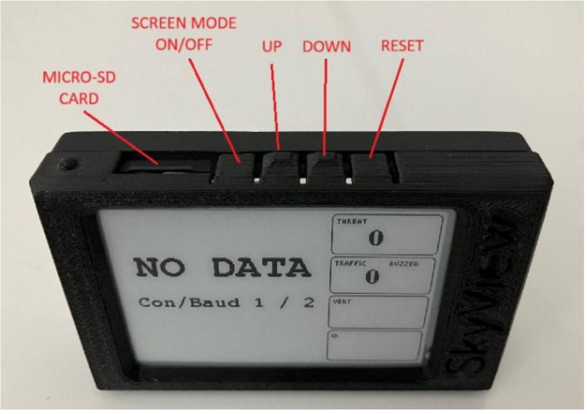


Illustration: The buttons

(Note: Con/Baud 1 / 2 shows the device is configured for connection serial port at baud 19200, but no data is being received)

Button	Name	Description
1	Mode	Long press = SkyView on/off. Short press = cycle through screen view modes.
2	Up	In Radar and Nav Views, radar zoom out. In Text_View, previous aircraft info (one page per aircraft)
3	Down	In Radar and Nav Views, radar zoom in. In Text_View, next aircraft info (one page per aircraft)
4	Reset	System reset. Avoid pressing this button. The system will be inoperable whilst the reset is in progress and until aircraft data is reacquired.

2.2 Connections

2.2.1 USB micro-B socket.

Used for software updates or powering the device only. SWD Version 3 is not capable of data in/out by this socket as this capability was not present at the time of code branching.

2.2.2 Side socket

Connection point for data from/to conspicuity transceiver using TTL level signals. Do not apply RS232 or USB level signals as they may permanently damage the PCB. Also power connection pins.

The WAGA Flarm has a TTL level data and 3.3V or 5V power output that can be directly hardwired to this socket.

2.2.3 Rear socket

Used to connect speaker and/or buzzer. The SWD has a built-in buzzer as standard.

3 Description of display screens

3.1 Screen view modes

There are three screen ‘views’ that can be cycled through using the left (landscape) or top (portrait) button. The screen view mode for start-up is configured and saved via the WebUI.

3.1.1 Radar_View

Intended as the normal view used by glider and tug pilots. The screen displays a square Radar Panel and four NavBoxes for information. The host aircraft is at the centre of the radar. The zoom level of the radar is shown in the bottom left corner of the radar and can be changed at the centre.

3.1.2 Nav_View

Intended to aid tug pilots undertaking paddock retrieves. Can also be used for glider pilots for emergency navigation back to their base. The Radar Panel performs as per the Radar View but with the addition of an RMI needle, and the NavBoxes display navigational data.

3.1.3 Text_View

Provides detail of aircraft detected. This view is retained from the original SoftRF software but is unlikely to be used often.

3.2 Screen layout and view mode contents

The screen layout orientation is set as Portrait or Landscape via the WebUI. It cannot be changed using the buttons.

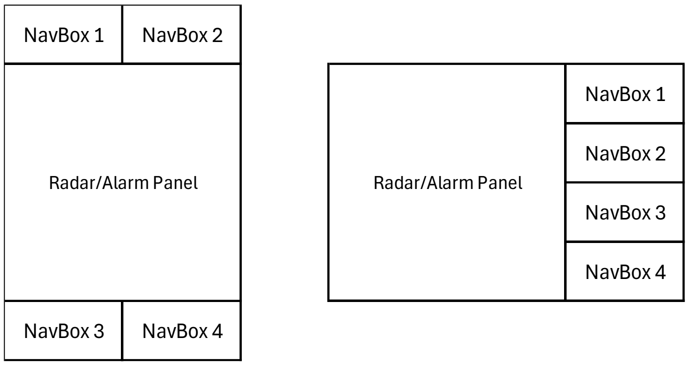


Diagram: Portrait and Landscape layouts

3.3 Radar Panel

The radar like panel will only show traffic that has been detected using EC interaction. Whilst it is referred to in this document as the Radar Panel, it does not use conventional radar detection (ie radio waves reflected by the reflected) and thus may not show all aircraft in the proximity.

The Radar Panel is a square area of the screen used to show detected traffic and, in the event of a FLARM Alarm, the Radar Panel is replaced with the Alarm Panel.

ThisAircraft is always shown as a small aircraft icon in the centre of the Radar Panel.

The Radar Panel can be configured as North-Up or ThisAircraft Track-Up:

- In Track-Up mode, the ThisAircraft icon always points up and the current track is shown at top centre of the Radar Panel.
- In North-Up Mode, 'N' is shown at top centre of the Radar Panel and the ThisAircraft icon will be rotate to an angle according to its track bearing.

It is recommended that Track-Up is used as this is the easiest to comprehend where other aircraft are located relative to the pilot sitting in the ThisAircraft.

The Radar Panel scale/zoom level may be set at 4 different levels and has metric and imperial options. There is always a large circle shown with radius matching the current scale setting as stated bottom left of the Radar Panel.

Proximate traffic is displayed as a standard target icon which is an elongated triangle that points in the direction of the traffic's heading. A small + or – beside the triangle indicates if the aircraft is above or below ThisAircraft's altitude. No '+' or '–' means the traffic is approximately at the same altitude as ThisAircraft.

From version WAGA02, the Comp ID is shown at 4 o'clock beside the traffic icon. If there is no SD card, no OGN database loaded or the aircraft is not in the database, then the last 3 digits of the Flarm's 6-digit ID code will be displayed.

If ThisAircraft's EC transceiver passes the required data to the SWD then:

- Tugs are shown with a single circle around the standard icon.
- ADSB derived traffic information is shown with 2 circles around the standard icon. This traffic is likely to be GA aircraft*.
- Mode-S traffic direction and altitude cannot be determined so is shown as a circle of dots at a radius corresponding to the approximate range.*

*Requires FLARM data port Protocol 9 or higher.

A line is drawn from the centre of the radar to the Nearest aircraft or the aircraft causing an alarm.

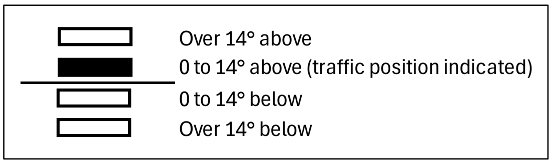
3.4 Alarm Panel

When Flarm alarms are detected, the Radar Panel is replaced by the Alarm Panel.

The Alarm Panel provides the 'where to look' information in a simple format that can be read in a glance so that the pilot is directed where to lookout for traffic that is a collision threat.

The information provided is:

- A lookout relative bearing indicator pointing in the direction of the threat and comprising 1, 2 or 3 black triangles depending on alarm level. Note that this indication will always be relative to ThisAircraft's current heading even if the Radar Panel is set to North-Up.
- A relative altitude angle indicator comprising 2 bars above the horizon and two below the horizon. One bar is all black indicating where to look vertically. The bars mean:



3.5 NavBoxes

NavBoxes are used on the Radar_View and Nav_View screens as follows:

NavBox	Radar_View Mode	Nav_View Mode
1	Traffic: Number of aircraft detect or Flarm Alarm Level	
2	Nearest: Distance to Nearest or Threat	Brg: Bearing to waypoint & Waypoint selected.
3	Vert: Threat relative height	Dist: Distance to waypoint
4	ID: Comp ID or Flarm ID	GS: Ground Speed

Note: WAGA03 NavBox use is slightly different to WAGA01

3.6 Audio sound output

Audio sound output to a speaker or pilot headset is possible when Sound is set to VOICE or TONE, and the appropriate .wav sound files are present on an installed micro-SD card.

VOICE Messages are composed as a sentence comprising words for: alert warning level, where, range and elevation depending on the traffic. Alarms are shorted than advisories.

TONE This setting plays a specific .wav file on the SD card for each of: Traffic advisory, Alarm level 1, Alarm level 2 and Alarm level 3. Whilst it was envisaged these would be short bleeping tones to gain the pilots attention, they can be voice messages. The owner can use their own .wav file provided the standard file names are used: Tone_Alert, Tone_Alarm1, Tone_Alarm2 and Tone_Alarm3.

Voice and Tone announcements delay the indication of traffic alarms on the Alarm Panel by up to 1.4 seconds. The Buzzer causes no delays.

There is no volume control in the SkyView configuration setting.

3.6.1 Audio speaker

An external 4-ohm speaker can be connected to the SkyView rear socket or the solder terminals as shown at Annex A. An 8-ohm speaker will normally work satisfactorily and not cause harm to the SWD but will deliver less power. Polarity is not usually an issue for a speaker.

3.6.2 Audio connection to aircraft radios

For audio to a tug or motorglider pilot's headset, the speaker can be replaced with a direct connection to the aircraft radio. Most GA radios have a connection point for an audio or intercom input so consult the manufacturers installation manual. A preset potentiometer is likely to be required to set the maximum gain and a potentiometer accessible to the pilot for headset volume control.

Polarity must be considered if connecting directly to other aircraft systems. Impedance matching is not likely to be critical but an impedance matching/isolating transformer is often the best solution for eliminating 'ground loop' interference noise.

If pilots use Bluetooth equipped headsets, then a Bluetooth module could possibly be added to SkyView to enable audio connection. This would need to be a different Bluetooth module to the built-in data Bluetooth.

4 Configuration

Configuration of the SkyView settings is required during installation or if a change is required. The settings are saved in memory so whenever the SkyView is started, it reads and applies these settings.

Configuration is done via the SkyView built-in local internet host using a Web User Interface (WebUI) usually on a smart phone. There is no requirement to connect to an internet router or smartphone hotspot with worldwide web access. This does not require a special application on the smart phone since it is merely acting as a browser and opening a HTML page received from the SkyView. This works on Android and iPhone.

4.1 Accessing the WebUI

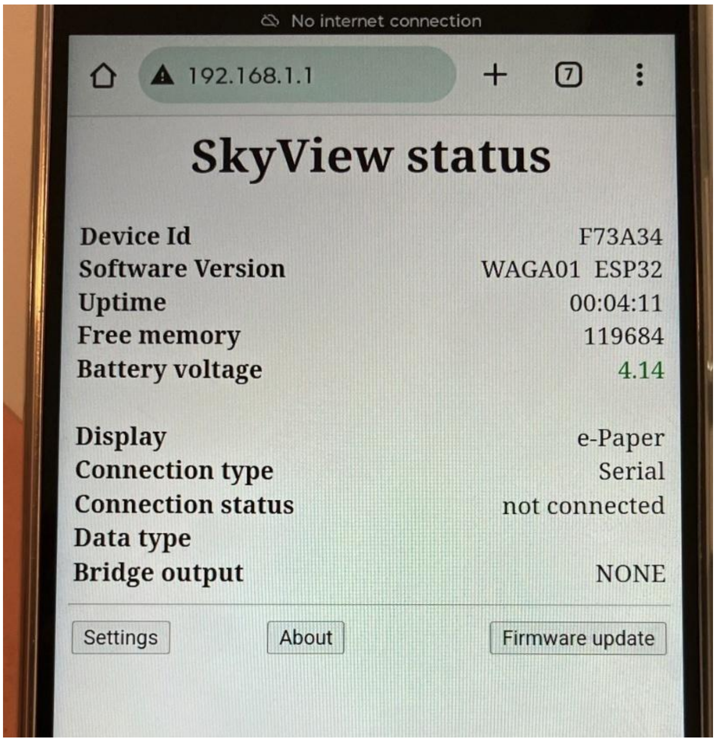
4.1.1 Making the connection Smartphone to SkyView

Power up the SkyView. Using a smartphone or PC 'WiFi Connections', identify the SkyView device which should be visible in the format 'SkyView – abc123'. Connect using the code/password 12345678.

Then use a smartphone web browser to search for and display IP address 192.168.1.1. Tips:

- It may be necessary to clear the browser cache before browsing.
- It may be necessary to ignore security warnings before this IP address will load.

The SkyView acts as an internet host and will send the WebUI 'SkyView status' page shown below to the smartphone browser screen.



Settings - Settings menu with options as described below.
About - about the SkyView software and a credits list of contributors.
Firmware update - not tested for SWD

Note the SkyView has a Device ID in same format as a Flarm or ICAO ID, but this is not disclosed to other traffic or used for any significant purpose other than identification from the list or available WiFi sites. It identifies your SkyView if there are others in range when you go to login to the WebUI.

4.2 WebUI Settings Menu and options

Setting	Setting options
Display Adapter	e-Paper TTG T5S ← use this for SWD e-Paper Waveshare ESP32
Connection type	**Serial** - wired connection. ← default for SWD. **WiFi UDP** - WiFi wireless connection **Bluetooth SPP** - 'Classic' Bluetooth wireless connection. Disabled in WAGA01. **Bluetooth LE** - Bluetooth LE connection (HM-10 compatible).
Protocol	**NMEA** - NMEA-0183 with (some) FLARM extensions **GDL90** - Garmin GDL90 .
Baud rate	of Serial connection. 4800 9600 **19200** ← Default rate for most FLARM devices 38400 57600 115200 2000000
Source ID	For a WiFi connection - SSID of WiFi UDP NMEA or GDL90 data server (Access Point) ; For a Bluetooth connection - advertised 'Name' of a Bluetooth device you want to connect to. Typically, the "server" is a PowerFlarm but can be other sources such as SoftRF Edition, SkyEcho, PilotAware, Stratux or similar equipment.
Key	For a WiFi connection - WiFi AP PSK key; Usually ‘**12345678**’ For a Bluetooth connection - 'Pin' of the Bluetooth device mentioned above. * *- only Bluetooth 'Simple Pairing' (no key) method is currently supported.

Setting	Setting options
Bridge Output	Data received by the SkyView is resent (without filtering or processing) to the selected port for sharing with other devices **None** ← use this for SWD **Serial** **WiFi UDP** **Bluetooth SPP** **Bluetooth LE**
Units	**metric** - display and say information in metric system **imperial** - display and say information in imperial system **mixed** - almost the same as **metric**, but all the altitudes are in feet.
Screen layout	**Portrait** **Landscape** Radar Panel and NavBox functionality is identical but arranged in different positions.
View mode	An option to choose from three different display views. **Radar** - radar-like view. Intended default screen for gliders **Text** - plain text representation of traffic information **Nav** - radar-like view. Intended default screen for tugs You can change the view in runtime by pressing the **MODE** button.
Radar orientation	**Track Up** - ThisAircraft's track over the ground is up. SWD recommended. **North Up** - GNSS North is up
Zoom level	Initial value of radar panel zoom level (scale factor). **lowest** 5 nm or 10 km **Low** 2.5 nm or 5 km **medium** 1 nm or 2 km **high** 0.5 nm or 1 km You can change the zoom level in runtime by using the **UP** and **DOWN** buttons.
Nav Waypoint	3 waypoints in Western Australia. In SWD Ver 3 this is hardcoded in an array. These can be changed by modifying the source code. **YBEV** **YCUN** **YNRG**
Aircraft data	The preferred database for decoding unique aircraft's ID code. **Auto** - selection is to be made based upon aircraft's NMEA "ID type" value (when available) **FlarmNet** **GliderNet** ← SWD can only use OGN data in CDB format **ICAO** This option makes sense only when there is a valid memory card in SkyView's micro-SD slot.
ID preference	When an aircraft's record is found, "**Text view**" will show text identification of this aircraft as: **Registration** - registration number of this aircraft **tail/CN** - tail / competition number (when available) . ← Only valid option for SWD. **make & model** - make and model of the aircraft (when available) ; You can choose which particular field of the record will be displayed there. If no records are found, then the hexadecimal ID value will be shown.
Sound	Pilot is made aware of traffic alerts by voice or buzzer. **Off** - no voice, tone or buzzer traffic alerts **Voice*** - composite information voice messages. Speaker must be connected. **Tone*** - tones or set messages. Speaker must be connected. **Buzzer** – Only when a buzzer is connected *These options are only possible when there is a correctly loaded data card in the micro-SD slot. Buzzer does not delay display refresh. Voice and Tone delay screen refresh whilst message is sounded.

Setting	Setting options
e-Paper 'ghosts' removal	Periodic full e-Paper screen refresh. Typically takes 1-2 seconds. **off** - do not apply this full screen refresh **auto** - **2 minutes** - enforce full screen refresh every 2 minutes (counter against heavy 'ghosting' effect); **5 minutes** - enforce full screen refresh every 5 minutes **10 minutes** - enforce full screen refresh every 10 minutes (little or no 'ghosting') It helps to remove residual traces of previous information, so-called 'ghosts'. 'Ghosts' are known to be temperature and UV dependent.
Traffic advisories filter	**Off** - no filters, except no advisories for traffic with zero ground speed. **Altitude (± 500 metres)** - Advisories only issued +/- 500 metres relative to ThisAircraft's altitude. **Alarm Only** - Alarms but no advisory messages
Power save	**Disabled** - do not use any of power saving methods **WiFi OFF (10 min.)** - turn SkyView WiFi AP off in 10 minutes after last client has closed its connection. Recommended.
Team Member Id	**Team Id** is a 6-digit hexadecimal unique identifier of your team member's device *(SoftRF, ICAO, FLARM, Skytrax, OGN, PAW, etc ...)* SkyView will depict this team member on the Radar Panel in a different manner. Not used by SWD.

4.3 Default settings

Illustration: Default settings for SWD firmware Version 3

4.4 Settings backup and restore

To **back-up** your settings - take a screenshot of the WebUI settings page.

To **restore** your settings - manually set all necessary selectors, buttons and knobs of the WebUI settings page while using earlier saved screenshot as a reference.

4.5 Sound files

The correct sound files must be continually present on the SkyView micro-SD card in a directory 'audio' directly off the drive root. It is recommended that owners keep a backup of the sound files. The initial batch of SWDs are supplied with a micro-SD card loaded with the voice files.

Instructions for obtaining the sound files can be found at:

https://github.com/lyusupov/SoftRF/wiki/SkyView.-SD-card

Voice .wav files must be in format: 16 Bit, sample rate 22050, Channel format LEFT only.

4.6 OGN Database file

This is optional. The SWD is able to lookup traffic IDs in an OGN database file held on its micro-SD card. It enables the display of FLARM Competition IDs, usually the last 2 or 3 letters of the registration (eg "DGZ'), rather than the unintelligible ID Hex code (eg 'ABC123').

The presence of an aircraft in the OGN database is dependent on the owner registering it, thus not all traffic will yield a Comp ID.

The initial batch of WAGA FLARMs have been registered on OGN and are included in the data file on the micro-SD card supplied with the initial batch of SWDs. If you wish to amend the registration consult, your Club representative for details of the http://wiki.glidernet.org/ddb account.

The OGN database was created for glider information exchange and thus ADSB information (eg for GA aircraft) is not normally found there.

It is recommended this file is updated regularly and whenever a new EC transceiver ID operating regularly in your proximity is identified. A daily updated copy of the OGN database in .cdb format file is available from:

http://soaringweather.no-ip.info/ADB/data/

The .cdb file should be loaded onto a micro-SD card, formatted FAT32, in a directory 'aircrafts' in the root directory. The micro-SD card must remain in the SkyView.

The micro-SD card is inserted with the connectors in the same direction as the display face. Click into the lock position.

5 Operational use

5.1 Start-up

If the SWD close down sequence was used, a 'sleep screen' will be on screen. Otherwise the last image displayed immediately prior to power shutdown will still be evident (this is a characteristic of E-Paper displays).

- Apply power to the SWD (and press the Mode Button to switch on if it has a battery).
- A startup screen which includes the software version will briefly appear.
- If Sounds is configured in the settings as VOICE, and a speaker or headphones are connected, then a male and then female voices saying "Notice" will be played to confirm functionality and aid setting volume.
- If Sounds is configured in the settings as TONE, and a speaker or headphones are connected, then the Tone_Alarm3.wav file will be played to confirm functionality and aid setting volume.
- If Sounds is configured in the settings as BUZZER, and a buzzer is connected, then 2 beeps will sound to confirm the buzzer is functioning.

5.2 Pre-flight checks

5.2.1 No Data

The SWD requires a valid data stream and a GPS fix from the conspicuity transceiver.

Following startup, the Radar Panel will briefly show ‘No Data’ and the SWD’s current connection type and baud rate settings (enumerated). For SWD, this is normally ‘1 / 2’, meaning ‘serial / 19200’.

If no data is available, this message will persist. Check that data is connected and the EC transceiver baud rate matches the SWD**.**

5.2.2 No Fix

The SWD requires a valid GPS fix for ThisAircraft. If no fix is available, the Radar Panel will show ‘No Fix’ and the current detected data protocol. For a SWD connected to a PowerFlarm, this should be ‘NMEA’.

If this message persists, it usually means the GPS has not been able to get a fix. Ensure the EC transceiver GPS antenna is not obstructed from satellite signals (eg in a hangar) and check the GPS antenna is properly connected. It usually takes 1 minute but can take longer, usually if the signal is poor. Eg. Installation near or under metal or carbon fibre

5.2.3 View mode selection

If a Fix is found, the default View Mode held in the Settings will commence. If a different view mode is required for the flight, use the Mode Button.

Glider pilots are likely to choose to fly with the Radar_View and tug pilots the Nav_View. Both screens share the same Radar Panel and FLARM Alarm Panel functionality.

5.2.4 Track-up / North-Up

Check whether the Radar is Track-Up (recommended) or North-Up. This can only be changed via the WebUI.

5.2.5 Radar Zoom level

At startup, the zoom level will be as per the settings. If a different zoom level is desired, use the two centre selection buttons.

Your SWD is now ready for flight.

5.3 Normal use: Radar_View

Monitor the SWD as part of your instrument scan.

You may wish to change the Radar Panel Zoom level using the Up/Down selection buttons as appropriate to your stage of the flight.

A summary guide explaining the screen displays is at Annex A.

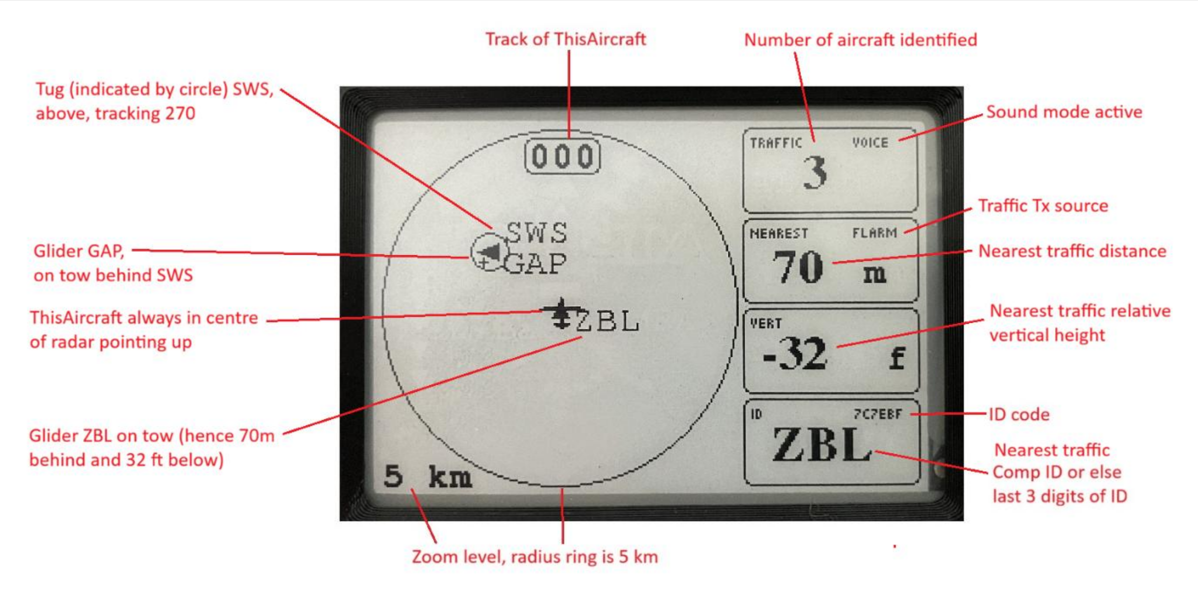


Illustration: SkyView fitted in a towplane. Radar_View with zoom level 5km radius to scale ring and track-up. ThisAircraft's track is 000 degrees. Nearest traffic is ZBL (a glider on tow). There is another tug (SWS) with a glider (GAP) on tow to the northwest above.

The Comp ID is shown at 4 o'clock to the icon for all aircraft, except tugs where it is shown 2 o'clock. This means both Comp IDs are visible when a tug/glider combination is conducting an aerotow launch or transit.

Aircraft on the ground with zero ground speed will not show their Comp ID so as to eliminate unreadable multiple IDs shown at the same place when gliders are lined up to launch.

Displayed directions relate to the ThisAircraft and traffic movement relative to ground (GPS) track.

If the Reset button is accidently pressed, it will take a few seconds before the SWD resumes normal operation using the configuration settings.

At the end of the flight it is best to shut down the SWD using a long press on the Mode Button. However, no harm will be done if power is simply cutoff. It is recommended the screen is covered when not in use to protect it from sun damage..

5.4 Traffic Advisory Alerts

Traffic advisory alerts are given when a new aircraft is identified within 6 km. If the aircraft remains within 6 km, the advisory will be repeated 5 minutes later.

Traffic Advisories can be limited to +/- 500m vertical altitude within 6km by setting Traffic Advisories Filter = Altitude (± 500 metres). The filter can only be changed via the WebUI configuration app, not recommended whilst in flight.

Traffic Advisories are stopped when setting Traffic Advisories Filter = Alarm only.

The configuration 'Sounds' setting determines how Traffic advisories are presented as follows:

Off. No sound.

Voice. A male voice will announce 'Traffic' and give its direction, how far and relative height. Eg. Traffic, 2 o'clock, 4 km, 12 hundred feet, above". Relative heights above +2000 or below -2000 are announced as "2000". For non-directional traffic (eg mode-S), 'direction' is spoken as "Aware".

Tone. Will play the Tone_Traffic.wav audio file on the SD card. This may be a tone or a word message.

Buzzer. No buzzer for advisories.

5.5 FLARM Alarms

If your EC transceiver can receive FLARM Alarms and send them to the SWD, then the SWD should alert the pilot to FLARM Alarm levels 1-3 by switching to the Alarm Panel in place of the usual Radar Panel.

FLARM Alarms are defined in FTD-012 and provide a time-based estimate until impact. In this manual, one second has been deducted from FLARM definitions to allow for the SWD processing time.

When a Flarm Alarm is encountered, only take a short look at the SkyView display and immediately look outside to find and identify the collision threat. Never try to look inside the cockpit or at the display when planning traffic avoidance manoeuvres.

If ThisAircraft is turning or thermalling when a FLARM Alarm is encountered, the ‘Look Out’ indicator will move around the radius ring showing the changing relative angle to the threat.

Note: There is no means of cancelling the alarms. The alarms will persist until the threat is no more.

When FLARM Alarms are received by the SWD, it assists the pilots as follows:

5.5.1 FLARM Alarm Level 1 = (14-19 seconds to impact)

When Sound = Voice is set, a female voice will announce an ALERT, eg “Alert, 6 o’clock, low”.

When Sound = Tone is set, the ‘Tone_Alarm1.wav audio file on the SD card will play. This may be a tone or word message.

When Sound = Buzzer is set, a single short buzz will sound and repeat at a slow rate as long as the threat continues.

The normal Radar Panel will be obscured and replaced with the Alarm Panel. One black triangles in a line from the centre of the radar point where to ‘look-out’ and that the alarm level is 1. The vertical angle indicator filled-in black shows where to look vertically with reference to the horizon.

On the Alarm Panel, a second black triangle will be added closer to the centre of the radar along the ‘look-out’ indicator line. It signifies the increased Alarm Level, not the position of the threat.

NavBox1 will show ‘ALARM 1’. The relative height above or below (-) is given in NavBox3. Inside the look-out black indicator will be shown + or – if the threat aircraft vertical relative height is more than 50ft above or below ThisAircraft.

5.5.2 FLARM Alarm Level 2 = (9-14 seconds to impact)

When Sound = Voice is set, a female voice will announce a WARNING, eg “Warning, 8 o’clock, level”.

When Sound = Tone is set, the ‘Tone_Alarm2.wav audio file on the SD card will play. This may be a tone or word message.

When Sound = Buzzer is set, a single short buzz will sound and repeat at a medium rate as long as the threat continues.

The normal Radar Panel will be obscured and replaced with the Alarm Panel. Two black triangles in a line from the centre of the radar point where to ‘look-out’ and that the alarm level is 2. The vertical angle indicator filled-in black shows where to look vertically with reference to the horizon.

NavBox1 will show ‘ALARM 2’.

5.5.3 FLARM Alarm Level 3 = (0-9 seconds to impact)

When Sound = Voice is set, a female voice will announce DANGER, eg “Danger, 11 o’clock, high”.

When Sound = Tone is set, the ‘Tone_Alarm3.wav audio file on the SD card will play. This may be a tone or word message.

When Sound = Buzzer is set, a single short buzz will sound and repeat at a fast rate as long as the threat continues.

The normal Radar Panel will be obscured and replaced with the Alarm Panel. Three black triangles in a line from the centre of the radar point where to ‘look-out’ and that the alarm level is 3. The vertical angle indicator filled-in black shows where to look vertically with reference to the horizon.

NavBox1 will show ‘ALARM 3’.

All FLARM Alarm warnings will cease when the FLARM Alarm level returns to zero. The threat assessed traffic will then become the one that is closest and be indicated with a line to its position on the Radar Panel.

5.6 Nav_View

Nav_View is intended for tow pilots conducting aero retrieves from paddocks but may be of use for gliders as an emergency navigate-to-base tool. It is selected using the Mode button.

Glider pilots who have landed out in a paddock, by standard convention, report their position as a bearing to and distance from the base airfield of the tug. With this information the towplane can fly to the glider's location.

The Nav-mode screen has textboxes that show the towplane's current position as bearing to and distance from the base, and the standard Radar Panel has a needle (like a Radio Magnetic Indicator RMI) that always points to the base waypoint. The towplane pilot uses this information to fly the outbound back bearing of the bearing reported by the glider.

Provided the glider has an operational Flarm, as the towplane approaches, the tow pilot should see the glider to be recovered as a traffic icon on the radar screen. The towplane's current ground speed is also displayed, enabling the pilot to quickly make a mental calculation of flight time to the glider. It will show the Comp ID beside the traffic icon despite it having no groundspeed.

If a Flarm Alarm is received, the Radar Panel will be obscured and the Alarm Panel shown. After the alarm has passed, the screen will revert to Radar_View.

The three WAGA airfields are hardcoded within the SWD software as navigation waypoints ie the tug base airfield. The airfield is selected using Setting 'Nav Waypoint'. Instructions for amending the SWD code for other locations are provided in the Github.

6 Installation

Installation and operation must comply with applicable local official regulations and requirements. It must be on the basis of non-interference or hazard to the existing suite of other equipment necessary for safe flying operation.

SkyView EZ and SWD do not have an ETSO or FAA-TSO airworthiness certification. Make sure that it is legal to install it in your aircraft.

6.1 The SkyView case

Install the SkyView where it can easily be seen. Readability largely depends on the viewing angle so make sure that the viewing angle is as perpendicular to the line of sight as is possible.

A SWD can be mounted in portrait (buttons on right) or landscape (buttons on top) orientation and the appropriate orientation setting selected in the config.

Plan sufficient space for connecting/disconnecting cables. The SkyView display should not be installed at a distance less than 30cm to a magnetic compass else it may contribute to compass errors.

The SkyView case/enclosure is not watertight. The ingress of solid particles or liquids is to be strictly avoided. If the unit gets moist, dry it before further use. If the unit gets wet, please consult an expert / repair facility to adequately clean the unit before further use.

SkyView displays do not contain a security glass front. Mechanical force applied to the display will destroy the display. The initial batch of SWDs were supplied with the manufacturer's protective plastic left in place. If this deteriorates, then remove it.

It is suggested that adhesive tape, glue or Velcro is used to secure the case. It may be possible to disassemble the SkyView to drill and bolt the case to a panel but be careful that bolt/screw heads do not physically interfere or cause electrical shorts on the circuit board.

When installing cables, make sure to comply with basic rules of cabling in aircraft regarding to EMI minimization. Wrong routing may disturb critical systems like e.g. the aircrafts radio system. Aviation grade wire is recommended.

Ensure that cables do not interfere with control movement or emergency procedures. In particular, the canopy jettison/release must not be constrained.

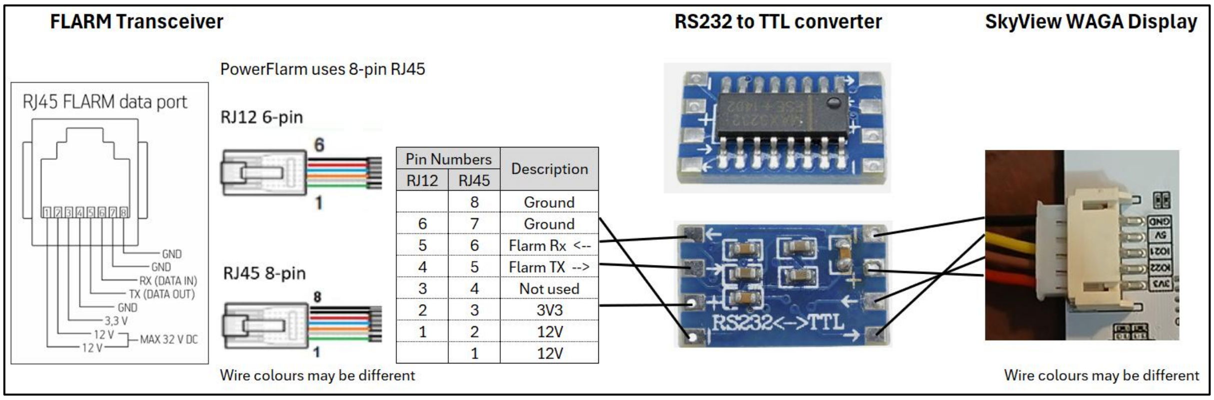
6.2 Power and data cabling for the SWD

If power is switched off (eg end of flight), fails without shut down of the SWD, no harm will be done but the E-Paper screen will freeze ie maintain the display that was current immediately prior to power loss.

Make sure that all requirements regarding power supply are being met. Using wrong polarity may permanently destroy the device. There is no internal fuse.

For the SWD, the usual point of connection for power and Rx data is the 5-pin connector on the long side. Then:

- When connected to a WAGA Flarm: Use the Flarm's special purpose permanently attached fly-lead which provides DC 3.3V power and TTL level data signals. No conversion of the data signal format or DC voltage power is required.
- When connected to any other PowerFlarm device via its RJ12 or RJ45 socket output:
  - Power: Unfortunately the Flarm "3V" power from the PowerFlarm is often insufficient to power the SWD which needs at least 3.3V. The symptom is the SWD continually reboots itself. In this case use the PowerFlarm 12V output and a 12V->5V DC power converter and connect via the SWD's 5V input pin.
  - Data: PowerFlarms output RS232 level data signals so to connect to the SWD 5-pin serial port, a RS232 to TTL converter is required. This is easily made using a small and inexpensive converter board. There are different formats of this type of converter board.



Pin Numbers		Description
RJ12	RJ45	
	8	Ground
6	7	Ground
5	6	Flarm Rx <--
4	5	Flarm TX -->
3	4	Not used
2	3	3V3
1	2	12V
	1	12V

Diagram RJ45 pinout, RS232 -> TTL lead

Shielded cable is suggested for long runs. The power draw is about 0.2A on average but can pulse on screen refresh and so require thicker long wires to avoid voltage drop. If your display resets or shuts down and gives a low battery message, then it likely needs a better power supply..

Connection of non-FLARM EC devices which provide data Tx/Rx via a USB socket, is possible directly to a SkyView display if it has enabling software code. SWD WAGA03 does not have an enabled USB socket for this purpose.

For clarity, the differences between TTL, RS232, USB data signal schemes are shown below.

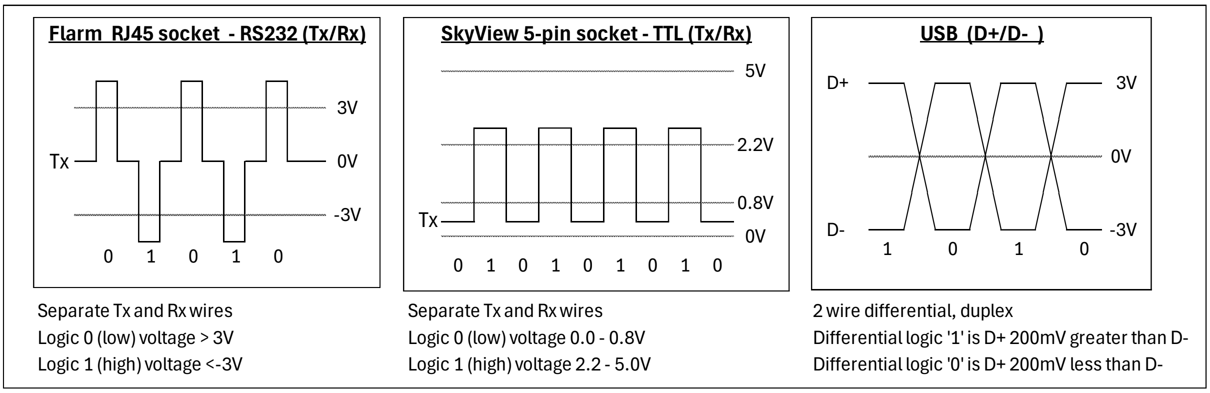


Diagram. The different data signal protocols

6.2.1 Rx Data and Power Via Rear Socket

It is possible to dispense with using the 5-pin socket on the long-side and instead connect power and Rx data via the rear 8-pin socket. This may be more convenient for some installations.

This requires a small coding change by a programmer and cannot be done via the WebUI.

The code change is in sketch: Platform_ESP32.h

```
Change:   #define SOC_GPIO_PIN_GNSS_RX  21
To:       #define SOC_GPIO_PIN_GNSS_RX  34
```

Then use pin IO34 for Rx and power pins as shown in diagram at end of the manual.

6.3 2x Displays in a tandem glider

2x SWDs may be installed in a tandem glider with hardwiring as follows:

- WAGA Flarm – wire the two SWDs in parallel to the WAGA Flarm's TTL data/power pigtails. They are connected in parallel to the board if provided by WAGA.
- PowerFlarm with 2x RJ45 sockets, provided both provide power – plug one SWD into each socket. Each SWD lead requires its own RS232<->TTL converter.
- Flarm with only one RJ45 or RJ12 socket providing power – using a simple 6P6C RJ45 Y-splitter. Only one RS232<->TTL converter is needed if fitted between the Flarm and Y-splitter.

Note: Only one of the SWDs should be allowed to transmit data to the EC transceiver. SWD Ver 3 does not transmit data but it may in future.

It may be possible to connect 2x SkyView displays using WiFi UDP or Bluetooth with the EC transceiver as the host. This has not been tested with the SWD and PowerFlarm.

7 Support

7.1 Software Updates

To update your display requires some knowledge of programming and uploading software.

Connect the SWD's USB micro-B port to a PC USB port and then use one of the following methods:

- Memory flashing using a flash download tool for the ESP32 and the binary files from Github. Instructions are on the Github.
- Loading the source code C++ files from Github into Arduino IDE which must then be compiled and downloaded to the SWD. See Github for IDE settings. Easy if you are familiar with Arduino IDE, otherwise, use the memory flashing method.

Source code can be found at: https://github.com/Bumpff/SoftRF_Skyview.

7.2 Technical Assistance

The SWD hardware and software is provided without warranty or guarantee of continuing support.

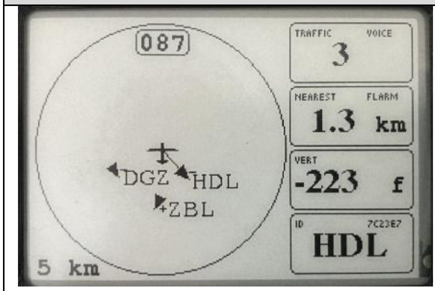
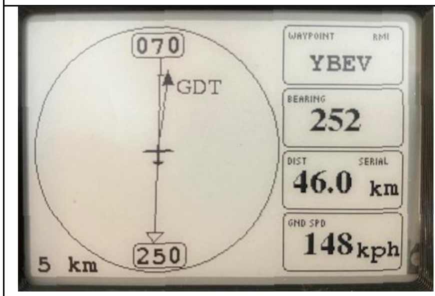
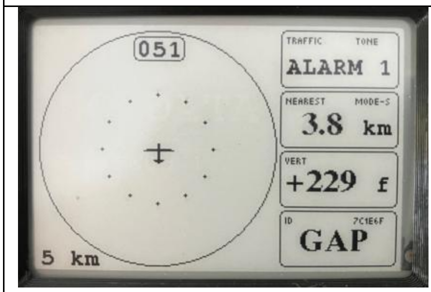
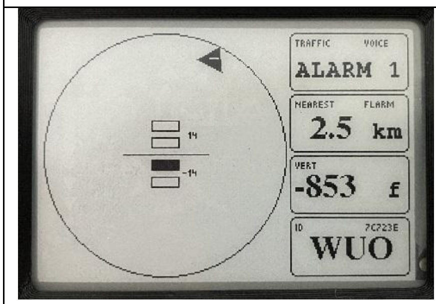
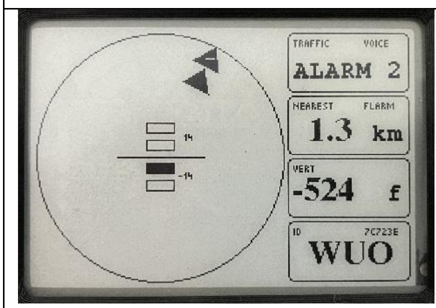
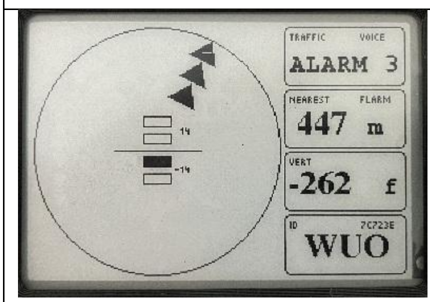
WAGA members using a SWD should contact their Club designated 'WAGA Flarm and SWD Super User'.

No technical assistance is provided for any software version other than the WAGA released versions. Support for non-WAGA members may be limited or not at all.

8 Further information

Flarm FTD-012 : Data port interface control document

Flarm FTD-014 : Flarm configuration specification

<table>
<tr><th>Radar Panel Examples</th><th>Description</th></tr>
<tr><td>
</td><td>
Radar_View mode with Track-Up. Scale ring is 5 km radius.
This Aircraft is the aircraft icon in centre of Radar Panel and current track is 087°. 3x traffic detected and shown with Comp IDs. +/- indicates above or below ThisAircraft.
Line from centre of radar to one of the traffic indicates traffic assessed as highest threat (no Alarms, so is the closest).
Nearest aircraft is ZBL 1.3 km, heading away from ThisAircraft, 223 ft relative height below.
</td></tr>
<tr><td>
</td><td>
Nav_View mode with Track-Up. Waypoint is set as YBEV.
The glider GDT reported its position as 072°, 49 km.
This aircraft: has Heading 070°, is 46.0 km from YBEV, the direction to YBEV is 252°, has ground speed 42 kph.
The RMI needle is pointing towards YBEV at 252°.
The nearest traffic is GDT below at approx 3 km, slightly right of track. This is the glider to be retrieved.
</td></tr>
<tr><td>
</td><td>
Radar_view mode Alarm Panel, Mode-S traffic.
This Aircraft has heading 051°.
Alarm state 1, 14-20 seconds until collison.
Current threat is ‘non-directional’ Mode-S traffic at distance 3.8 km, 229 feet above. The direction of the traffic is unknown and could be anywhere at the range shown by the circle of dots.
</td></tr>
<tr><td>
</td><td>
FLARM Alarm 1. 14-20 seconds until collision.
Look-Out Indicator (single black triangle just inside circle) directs pilot to lookout at 1 o’clock. Threat is 2.5 km away.
Vertical angle indicator directs lookout to below horizon 0-14°. +/- in black triangle also indicates threat is below.
Buzzer, Tone or female voice. Eg “Alert.1 o’clock, above”.
Threat has Comp ID ‘WUO’.
</td></tr>
<tr><td>
</td><td>
FLARM Alarm 2. 9-14 seconds until collision.
Look-Out Indicator (two black triangles just inside circle) directs pilot to lookout at 1 o’clock. Threat is 1.3 km away.
Vertical angle indicator directs lookout to below horizon 0-14°.
Buzzer, Tone or female voice. Eg “Warning.1 o’clock, above”.
Threat has Comp ID ‘WUO’.
</td></tr>
<tr><td>
</td><td>
FLARM Alarm 3. 0-9 seconds until collision.
Look-Out Indicator (three black triangles just inside circle) directs pilot to lookout at 1 o’clock. Threat is 447 m away.
Vertical angle indicator directs lookout to below horizon 0-14°.
Buzzer, Tone or female voice. Eg “Alert.1 o’clock, above”.
Threat has Comp ID ‘WUO’.
</td></tr>
</table>

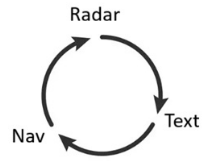
ANNEX B to SkyView WAGA Display manual : LilyGo T5S diagrams

Buttons location

Buttons function

MODE

Short press and release - change view mode



Long press (>2 sec) and release - soft power off

UP

Radar/Nav view - zoom out
Text view - previous aircraft info

DOWN

Radar/Nav view - zoom in
Text view - next aircraft info

RESET